Proof:

(i) TA = TP (hypotenuse) and $\angle TSA = 90$

TS is common (leg)

Hence, by RHS congruence, $\triangle TAS \equiv \triangle TPS$

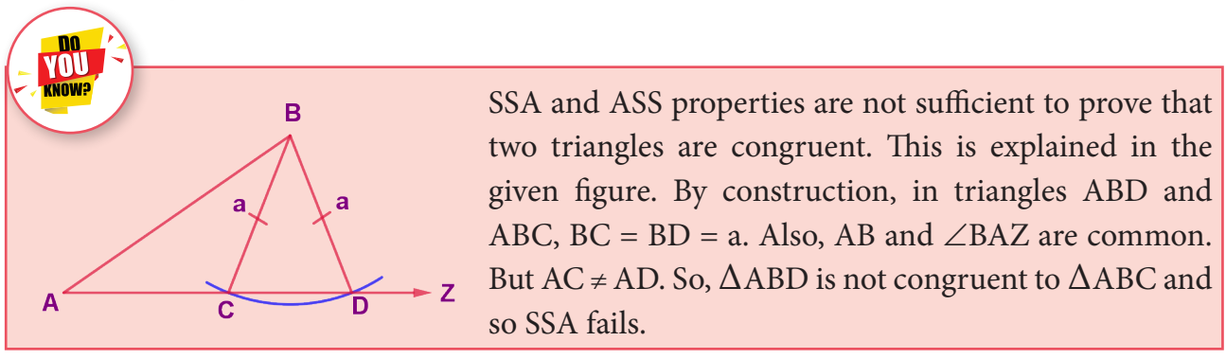
(ii) Given TA = TP

$\therefore \quad \angle P = \angle A$ (if angles then sides)

(iii) From (i) $\triangle TAS \equiv \triangle TPS$,

By CPCTC

$AS = PS$



SSA and ASS properties are not sufficient to prove that two triangles are congruent. This is explained in the given figure. By construction, in triangles ABD and ABC, BC = BD = a. Also, AB and ∠BAZ are common. But AC ≠ AD. So, ΔABD is not congruent to ΔABC and so SSA fails.

5.2.2 Similar Triangles

Consider two given triangles PQR and ABC. They are said to be similar ($\sim$) if their corresponding angles are equal and corresponding sides are proportional. That is $\angle P = \angle A$, $\angle Q = \angle B$ and $\angle R = \angle C$ and also $\frac{PQ}{AB} = \frac{PR}{AC} = \frac{QR}{BC}$. This is denoted as $\Delta PQR \sim \Delta ABC$.

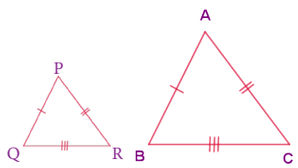


Fig. 5.11

There are 4 ways by which one can prove that two triangles are similar.

(i) AAA (Angle – Angle – Angle) or AA (Angle – Angle) Similarity

Two triangles are similar if two angles of one triangle are equal respectively to two angles of the other triangle. In the given figure, $\angle A = \angle P$, $\angle B = \angle Q$. Therefore, ΔABC $\sim$ ΔPQR.

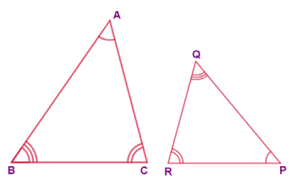


Fig. 5.12

(ii) SAS (Side – Angle – Side) Similarity

Two triangles are similar if two sides of one triangle are proportional to two sides of the other triangle and the included angles are equal. In the given figure, $\frac{AC}{PQ} = \frac{AB}{PR}$ and $\angle A = \angle P$ and hence ΔACB $\sim$ ΔPQR.

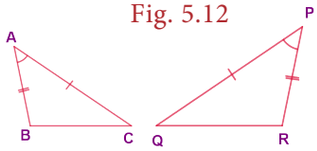


Fig. 5.13

8th_Maths_English Medium_Chapter 5.indd 161 23/11/2022 15:49:06